Supplementary Information for "Large Language Model Powered Knowledge Graph Construction for Mental Health Exploration"

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1 The Integrated Biomedical Databases

In this part, we delve into a comprehensive exploration of a wide array of existing biomedical databases. These databases are thoughtfully categorized into chemical-centric, pathway-centric, gene-centric, and Phenotype/disease-centric databases.

1. Chemical-centric database

- **Side Effect Resource (SIDER)**[1]: contains information on marketed medicines and their recorded adverse drug reactions. The information is extracted from public documents and package inserts. The available information includes side effect frequency, drug, and side effect classifications, as well as links to further information, for example, drug–target relations.
- **DrugBank**[2]: a bioinformatics and cheminformatics knowledge base with rich drug data information (i.e., chemical, pharmacological, and pharmaceutical) and comprehensive drug target information (i.e., sequence, structure, and pathway).
- **Comparative Toxicogenomics Database (CTD)**[3]: a regularly updated, chemical-centric resource that curates over 3.4 million evidence-based interactions involving chemicals, genes, phenotypes, anatomy, diseases, and exposure across multiple species, drawn from chemical-centric manually curated articles.

2. Pathway-centric database

- **Reactome**[4]: provides manually curated molecular details across various physiological and pathological biological processes in humans, including hereditary and acquired disease processes.

3. Gene-centric database

- **Gene Ontology (GO)**[5] [6]: focuses on defining gene activity on the molecular level, linking genes to locations in the body where its function is performed, and establishing links between genes and biological processes.
- **Bgee**[7]: a database for retrieving and comparing gene expression patterns across multiple animal species. It provides an intuitive answer to the question "Where is a gene expressed?" and supports cancer, agriculture, and evolutionary biology research.
- **ClinVar**[8]: a public repository that collects, stores, and displays information on human genetic variants and their observed relationships to phenotypes, including clinical significance assertions, supportive evidence, and review status, with data sourced from clinical testing, research, and literature reports.

4. Phenotype/diseases-centric database

- **DisGeNET**[9]: one of the largest available collections of genes and variants involved in human diseases. It integrates data from expert-curated repositories, GWAS catalogues[10], animal models, and the scientific literature. DisGeNET data are homogeneously annotated with controlled vocabularies and community-driven ontologies. Additionally, several original metrics are provided to assist the prioritization of genotype–phenotype relationships.

- **Human Phenotype Ontology (HPO)**[11]: a structured database of over 13,000 medically relevant phenotypes and disease-phenotype annotations, derived from clinical data. Its structure is a directed acyclic graph connected by *is-a* (*subclass-of*) relationships.
- **Mondo Disease Ontology (MONDO)**[12]: a semi-automatically constructed ontology that merges multiple disease resources to yield a coherent merged ontology. Unlike generic ontology mappings, MONDO employs rigorous semantics to distinguish when two disease names or identifiers are exactly equivalent versus merely being related. This meticulous distinction ensures a high level of clarity and precision in disease representation and inter-resource mappings.

5. Anatomy-centric database

- **Uberon**[13]: an integrated cross-species ontology consisting of over 6,500 classes representing a variety of anatomical entities, organized according to traditional anatomical classification criteria. The ontology represents structures in a species-neutral way and includes extensive associations to existing species-centric anatomical ontologies, allowing the integration of model organism and human data.

6. The Unified Medical Language System (UMLS)[14]: designed by the US National Library of Medicine, is a key resource that includes a Metathesaurus with controlled biomedical vocabularies. Concepts within this Metathesaurus are sorted into 127 semantic types, such as organisms and chemicals, linked by 54 relationships like the hierarchical "isa". The semantic network catalogs these and includes five major non-hierarchical relationship categories, capturing potential connections between instances of different types. UMLS aids in medical informatics with natural language processing, free access, and quarterly updates.

2 Entity and Relationship Types

Through an in-depth analysis of high-impact PubMed articles, we identify ten key themes in mental health research, focusing on prevalent mental disorders like depression, anxiety, and schizophrenia. These themes include pathogenesis, pharmacological interventions, therapeutic approaches, health-related factors, neurotransmitter and neural signaling dynamics, neural development and neurobiology, advances in neuroimaging and neurophysiology, genetic influences and polymorphisms, intersections of endocrinology and metabolic processes, along with immunological and inflammatory responses. We also pinpoint nine entity types (Health Factors, Anatomy, Phenotype, Sign, Method/Procedure, Physiology, Gene/Protein, Disease/Symptom, Drug/Chemical) and seven relationship types (Risk relationships, Diagnostic relationships, Characteristic relationships, Association relationships, Therapeutic relationships, Hierarchical relationships, and Abbreviation) that are central to mental disorders research. These categories offer a structured lens for examining the complexities of mental health, thereby enhancing the comprehensiveness of our MDKG.

2.1 Entity Types

1. **Health Factors**: This refers to factors that influence a person's likelihood of developing health issues, such as health behaviors, clinical care, social and economic factors, and physical environment factors.
2. **Anatomy**: This refers to bodily structures, including organs, tissues, cells, and other components within living organisms.
3. **Phenotype**: This refers to observable or measurable changes in an individual's anatomy, physiology, or signs.
4. **Sign**: This refers to measurements or indicators of the body or organ's functions or states, often obtained through laboratory tests or examinations.
5. **Method/Procedure**: This refers to techniques or procedures used in research, diagnosis, treatment, or prevention of diseases, as well as disciplines and occupations related to the biomedical field.
6. **Physiology**: This refers to biological, molecular, and cellular functions, including mental processes and organ or tissue functions.
7. **Gene/Protein**: This refers to a segment of DNA that carries genetic information involved in the sequencing of molecules and the formation of proteins, enzymes, and receptors.
8. **Disease/Symptom**: This refers to various types of abnormalities or dysfunctions, including congenital, acquired, injuries, pathologies, mental or behavioral issues, and cellular or molecular issues.
9. **Drug/Chemical**: This refers to substances with specific structures and functions, including organic and inorganic compounds, pharmacological substances, hormones, and hazardous materials.

2.2 Relationship Types

1. **Risk relationship**: This relationship refers to a factor that increases the likelihood of developing a specific disease or a factor that exacerbates the progression of a specific disease. For instance, poor dietary habits could increase the risk of heart disease, while smoking could exacerbate lung cancer progression.
2. **Diagnostic relationship**: The diagnostic relationships in our knowledge graph are categorized into two main types. The first type encompasses disease diagnosis methods, which include tools and techniques like psychological assessments, neuroimaging technologies, and biochemical tests. The second type involves identifying specific features or biomarkers that assist physicians or researchers in diagnosing particular diseases or pathological conditions. These features or biomarkers are typically characterized by their high diagnostic specificity and sensitivity, enabling effective differentiation between healthy and diseased states or among various disease states.
3. **Characteristic relationship**: Characteristic relationships in the graph describe the symptoms, clinical manifestations, or distinct features of a disease. These characteristics span multiple dimensions, including physiological, biochemical, behavioral, and genetic levels. For example, in the context of depression, characteristic relationships might encompass symptoms like persistent sadness, biochemical

factors such as serotonin imbalance, behavioral signs like loss of interest in activities, and genetic predispositions.

4. **Association relationship**: This relationship describes an association or connection between two or more entities, where changes in one entity may correspond with changes in another but not necessarily imply a cause-and-effect relationship.
5. Therapeutic relationship: This relationship type highlights the link between various treatments or interventions and their impact on a specific disease or condition. It includes a range of treatment methods like medication, psychotherapy, lifestyle modifications, surgical procedures, and others.
6. **Hierarchical relationship**: This relationship refers to the classification and hierarchical structure within the same entity type. For instance, within the 'Disease' category, major classifications like 'mental disorders' could be further divided into subclass diseases such as depression, anxiety, and schizophrenia.
7. **Abbreviation relationship**: This relationship type refers to the use of abbreviations or acronyms to represent longer terms or phrases, especially in medical or technical fields. For example, 'AIDS' is an abbreviation for 'Acquired Immune Deficiency Syndrome', and 'MRI' stands for 'Magnetic Resonance Imaging'.

3 Confidence Score Calculation

The confidence score integrates multiple quality indicators of scientific statements (triplets) using a weighted framework. Each component score is normalized to [0,1] and scaled by a predefined weight, ensuring the final score remains within the range [0,3]. Prior research on assessing study quality has highlighted key factors influencing a study's credibility and impact, including impact factor, statement frequency, and the number of distinct publication years[15–17]. Our framework incorporates these established factors and further refines their contributions based on insights from domain experts, ensuring a balanced and context-aware evaluation. The confidence score is computed as follows:

$$\begin{aligned} S_{confidence} = S_{prediction} + 0.8 \cdot S_{IF} + 0.6 \cdot S_{Frequency} + 0.4 \cdot S_{years} + 0.2 \cdot S_{date} \\ - 0.5 \cdot S_{contradiction} \end{aligned}$$

- **Model Prediction Score ($S_{prediction}$)**: Quantifies the base trustworthiness of the triplet generated by the prediction model. The model directly outputs the score, ranging from [0, 1].
- **Impact Factor Score (S_{IF})**: Reflects the academic influence of the resource.

$$S_{IF}(i) = 1 - \exp\left(-\alpha \mathrm{i}\right)$$

where i is the average impact factor of the source journal. To align with the 75th percentile impact factor value (6.6), the decay coefficient $\alpha = 0.25$ ensures $S_{IF} \geq 0.8$ when $i \geq 6.6$.

- **Frequency Score ($S_{Frequency}$)**: Measures how frequently the triplet appears across different resources.

$$S_{Frequency}(n) = 1 - \exp\left(\alpha(n-1)\right)$$

where n is the occurrence count of the triplet. With the 75th percentile $n = 4, \alpha = 0.55$ ensures $S_{Frequency} \geq 0.8$ when $n \geq 4$.

- **Distinct Publication Years Counts Score (S_{years})**: Captures the historical consistency of the triplet across years.

$$S_{years}(n_t) = 1 - \exp(-\alpha n_t)$$

where n_t is the number of distinct publication years. For the 75th percentile $n_t = 4, \alpha = 0.4$ yields $S_{years} \geq 0.8$ when $n_t \geq 4$.

- **Max Publication Year Score (S_{date})**: Rewards recency of the triplet's latest publication.

$$\begin{cases} S_{date}(t) = \exp(-\alpha(T-t)), & t \leq T \\ S_{date}(t) = 1, & t > T \end{cases}$$

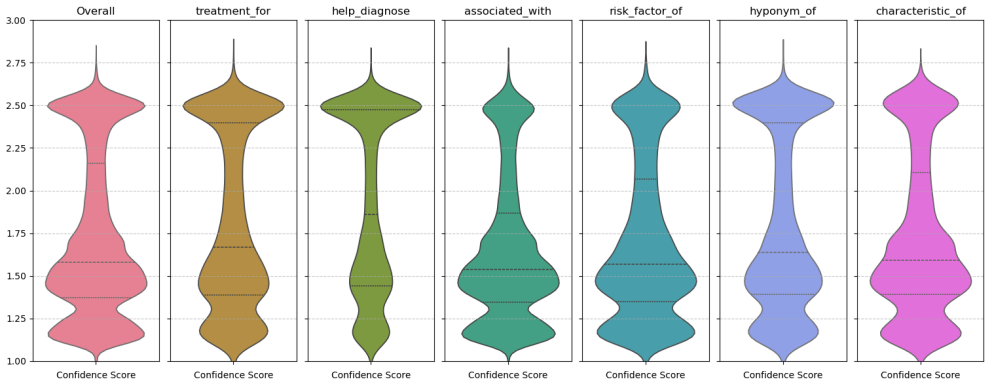
where T is the reference year (e.g., 2025) and t is the latest publication year. The Decay Rate $\alpha = 0.1$ imposes gradual penalties for older publications.

- **Contradiction Counts Score ($S_{contradiction}$)**: Penalizes conflicting statements from other sources.

$$S_{contradiction}(c) = \exp(-\alpha c)$$

where c is the count of contradictory triplets. $\alpha = 0.1$ ensures a penalty of 0.8 when $c = 2$.

In summary, the confidence score aggregates multiple aspects of a resource's quality and relevance, with each factor having a distinct role in shaping the final score. The decay functions ensure that the score is adjusted for diminishing returns, while the weights reflect the relative importance of each factor in the overall assessment. This approach allows for a nuanced evaluation of resources, balancing credibility, academic impact, historical significance, recency, and frequency of use. The distribution of confidence scores across different types of relationships is illustrated in Supplementary Figure 1.



Supplementary Fig. 1: The distribution of confidence scores for different relationships.

4 Additional Details of Information Extraction

4.1 Performance of Relationship Annotation with GPT-4

We used GPT-4 for pre-annotation, significantly reducing the time and effort required for manual annotation. GPT-4 performed two main tasks: (1) identifying noun phrases in abstracts and (2) determining whether a specific relationship (e.g., therapeutic or risk) existed between annotated entity pairs. The second task was treated as a binary classification problem.

We evaluated GPT-4's performance by comparing its annotations with manually reviewed data from 500 mental health-related abstracts (MDIEC). For entity identification, GPT-4 achieved an F1-score of 0.88, with a precision of 0.84 and a recall of 0.93, demonstrating strong recognition of relevant terms while occasionally including non-essential ones. For relationship extraction, performance was assessed against the manually reviewed dataset, with results summarized in Supplementary Table 1.

Supplementary Table 1: Performance of GPT-4 in Relation and Conditional Statement Extraction Tasks

Relationship Types	**GPT-4 F1 Score**	**Sample Size**
Hierarchical	0.92	1166
Association	0.83	2686
Characteristic	0.81	700
Risk	0.85	1199
Therapeutic	0.83	1239
Diagnostic	0.91	589
Abbreviation	0.97	1112
Conditional Statement Types		
Occurs in	0.75	1225
Located in	0.90	524

These results indicate that GPT-4 achieves high performance in identifying well-defined relationships (F1-scores above 0.9) but shows reduced accuracy for more ambiguous categories, such as Characteristic, Risk, Therapeutic, and Occurs in. The strong performance in relationship extraction is largely attributable to our reformulation of the original multi-class task (where the model determined the most appropriate relationship among nine possible types) into a binary classification task (where the model was only asked whether a specific relationship existed). This reformulation reduced categorical complexity, improving model accuracy, but significantly increased the cost of model usage.

4.2 Performance of Triplet Extraction with Spert.pl and Refinement with GPT-4o

Triplet extraction was performed in two stages. First, a BERT-based classification model (Spert.pl) was fine-tuned on the MDIEC annotated dataset for named entity

recognition (NER) and relationship extraction (RE) and then applied to unannotated abstracts to extract triplets. In the second stage, GPT-4o was used to validate and refine the extracted triplets, improving their accuracy and reliability.

1. **The performance of Triplet Extraction**
   For the NER task, Spert.pl performed multi-class classification to identify and categorize potential entities. For the RE task, the model determined whether a specific relationship existed between a given pair of entities. As shown in Supplementary Table 2, Spert.pl achieved strong performance in NER (macro F1-score: 0.88), but relationship extraction remained challenging (macro F1-score: 0.67). When evaluating the joint NER and RE performance—assessing whether both entities and their relationships were correctly identified—the model achieved a macro F1-score of 0.63.
2. **The performance of Triplet Validation**
   To enhance the accuracy of extracted triplets, GPT-4o was employed for validation and refinement. After this process, the micro F1-score for triplet extraction improved to 0.79, with micro precision increasing to 0.84. This substantial improvement significantly enhanced the quality of the knowledge graph while maintaining cost efficiency.

Supplementary Table 2: Performance metrics for NER, RE, and Joint NER & RE tasks

Task	Micro-P	Micro-R	Micro-F1	Macro-P	Macro-R	Macro-F1
NER	0.89	0.90	0.89	0.88	0.88	0.88
RE	0.69	0.67	0.68	0.68	0.67	0.67
Joint NER & RE	0.62	0.65	0.63	0.62	0.65	0.63
Valid Joint NER & RE	**0.84**	**0.74**	**0.79**	**0.83**	**0.70**	**0.76**

4.3 Performance of Table Information Extraction

We have conducted a manual review to assess the performance of extracting baseline characteristics from PDF tables. This review involved evaluating the accuracy of the process and identifying the causes of any errors encountered. The results of this evaluation are presented in Supplementary Table 3. Out of the 302 total documents reviewed, 238 were PDFs, among which 72 contained target tables expected to be extracted for analysis.

The review identified two primary types of errors. The first, Tables Not Found, occurred in 12 documents. This issue was primarily caused by two factors: (1) in two instances, the system failed to detect tables because their titles spanned multiple pages, making it difficult to correctly associate the title with the corresponding data; (2) in three cases, the system struggled to identify tables due to unclear or missing keywords (in our method, we differentiate tables by locating baseline-related keywords). The remaining seven cases were attributed to less identifiable reasons, likely related to complex formatting or irregular data layouts.

Supplementary Table 3: Baseline Characteristics Extraction Error Analysis

Total number of documents	302
Total PDFs	238
PDFs with target tables	72
Error 1. Tables not found	
Reason 1: Issues with table titles across pages	2
Reason 2: Issues with keywords	3
Other Reasons	7
Error 2. The table found but the final answer is wrong	
Reason 1: Incomplete table	1
Reason 2: Header issues (headers not extracted)	3
Reason 3: Horizontal tables	1
Reason 4: Issues attributed to ChatGPT	5
a. Incorrect or incomplete grouping with multiple headers	
b. Inaccurate identification of categories such as education and employment	
c. Still provides answers when incorrect tables are input.	

The second error type, Table Found but Final Answer is Wrong, accounted for 10 documents. The primary causes included incomplete table extraction, missing table headers, issues with Horizontal table formats, or incorrect answers generated by ChatGPT.

4.4 Comparative Analysis of Large Language Models for Biomedical Information Extraction

We systematically evaluated the performance of four language models—GPT-4 (gpt-4), GPT-3 (text-davinci-003), Claude-3-opus, and ChatGPT (GPT-3.5-turbo-1106)—on two key biomedical natural language processing (NLP) tasks: named entity recognition (NER) and relationship extraction (RE). Both tasks were formulated as multi-class classification problems. To benchmark performance, we used MDIEC, a dataset of 500 manually annotated medical abstracts from our institutional corpus. The dataset included nine entity types (Anatomy, Disease/Symptom, Drug/Chemical, Gene/Protein, Health Factors, Method/Procedure, Phenotype, Physiology, and Sign) and seven relationship types (Therapeutic, Diagnostic, Risk, Characteristic, Association, Hierarchical, and Abbreviation). A standardized few-shot learning framework was applied, and model performance was evaluated using precision, recall, and F1-score metrics.

For NER (Supplementary Table 4), Claude-3-opus demonstrated marginally superior performance to GPT-4 (F1: 0.87 vs. 0.85). In relationship extraction (Supplementary Table 5), GPT-4 achieved superior performance (F1: 0.66 vs. Claude-3-opus: 0.63). Both models significantly outperformed GPT-3.5-turbo (NER F1: 0.74; RE F1: 0.55) and GPT-3 (NER F1: 0.63; RE F1: 0.51) across all metrics.

Supplementary Table 4: Comparative Analysis of LLMs in Entity Recognition Task

Model	Precision	Recall	F1-score
Claude-3-opus	**0.89**	**0.85**	**0.87**
GPT3	0.67	0.59	0.63
ChatGPT	0.71	0.78	0.74
GPT-4	0.87	0.84	0.85

Supplementary Table 5: Comparative Analysis of LLMs in Relationship Extraction Task

Model	Precision	Recall	F1-score
Claude-3-opus	0.64	0.62	0.63
GPT3	0.45	0.40	0.42
ChatGPT	0.57	0.54	0.55
GPT-4	**0.67**	**0.64**	**0.66**

4.5 Comparative Performance of GPT-4 and Spert.pl in Named Entity Recognition Task

We evaluated the performance of Spert.pl (a BERT-based model) and GPT-4 on a named entity recognition (NER) task. Spert.pl was fine-tuned on 400 manually annotated biomedical abstracts for NER and relationship extraction tasks, while GPT-4 was assessed in a few-shot setting. Benchmarking was conducted on a held-out test set of 100 annotated abstracts distinct from Spert.pl's training data. The F1 scores across entity types are summarized in Supplementary Table 6.

Supplementary Table 6: Comparison of F1 Scores for Spert.pl and GPT-4 on Named Entity Recognition

Entity Type	**Spert.pl**	**GPT-4**	**Sample Size (n)**
Disease/Symptom	0.89	**0.94**	478
Gene	0.83	**0.92**	191
Health Factors	**0.82**	0.80	435
Method/Procedure	**0.90**	0.84	836
Anatomy	0.82	**0.93**	170
Drug/Chemical	0.88	**0.93**	180
Phenotype	0.78	**0.83**	86
Physiology	0.86	**0.89**	279
Sign	0.80	**0.83**	68

Note: Spert.pl was fine-tuned on 400 abstracts; GPT-4 was evaluated Few-shot. Test data comprised 100 non-overlapping annotated abstracts.

- **Performance on Well-Defined Entities**: GPT-4 demonstrated superior accuracy (F1 scores: 0.92–0.94) for entities with clear definitions, including Disease/Symptom,

Gene, Anatomy, and Drug/Chemical. This high performance is likely attributable to its extensive pretraining on diverse corpora, including biomedical datasets.

- **Ambiguous Entity Challenges**: Spert.pl achieved parity or outperformed GPT-4 on entities with ambiguous boundaries, such as Method/Procedure (F1: 0.90 vs. 0.84) and Health Factors (F1: 0.82 vs. 0.80), suggesting task-specific training data enhances resolution of contextually complex entities.
- **Data Dependency of Traditional Models**: The performance variability of Spert.pl (F1: 0.78–0.90) highlights the impact of training data quality and diversity, especially for low-sample entities such as Phenotype and Sign.
- **Practical Considerations**: Task requirements and cost-efficiency trade-offs should guide model selection.

4.6 The GPT Prompt

Listing 1: The GPT-4.0 Prompt for Noun Phrases and abbreviations Recognition

```
prompt = """
You should extract noun phrases and abbreviations from the text
    using a JSON format, for example: {"Noun phrases": [], "
    Abbreviations": {}}. The sentence is:
"""

example_Sent = """
    Outpatients with Major Depression Disorder (MDD, n=44) and
        healthy controls (HCs, n=28) were administered the Beck
        Depression Inventory-I (BDI-I), the Three-Dimensional
        Psychological Pain Scale (TDPPS), and the Beck Scale for
        Suicide Ideation (BSS), and then patients were allocated
        to two groups: suicide attempts (MDD-SA, n=12) and
        suicidal ideation (MDD-SI, n=32).
    """

example_answer = """
    {"Noun phrases": ["Outpatients with Major Depression Disorder
        ", "Healthy controls", "Beck Depression Inventory-I", "
        Three-Dimensional Psychological Pain Scale", "Beck Scale
        for Suicide Ideation", "Suicide attempts", "Suicidal
        ideation"], "Abbreviations": {"MDD": "Major Depression
        Disorder", "HC": "Healthy controls", "BDI-I": "Beck
        Depression Inventory-I", "TDPPS": "Three-Dimensional
        Psychological Pain Scale", "BSS": "Beck Scale for Suicide
        Ideation", "MDD-SA": "Major Depression Disorder - Suicide
        Attempts", "MDD-SI": "Major Depression Disorder - Suicidal
         Ideation"}}
"""

{"role": "system", "content": "You are a helpful assistant
    designed to output JSON."},
```

```
{"role": "user", "content": prompt + example_Sent},

{"role": "assistant", "content": example_answer},

{"role": "user", "content": prompt + Target sent}
```

Listing 2: The GPT-3.5 Prompt for Cohort Name Extraction

```
prompt = """
list the study population cohort name or biobank name in this
    study using a JSON form with "population name"
"""
```

5 Additional Details For Disease Prediction

Supplementary Table 7: The Excluded Phecodes for the Control Group (Phecodes 295-306.99)

Phecode	**Phenotype**
295.3	Psychosis
296	Mood disorders
300.1	Anxiety disorder
295.1	Schizophrenia
296.22	Major depressive disorder
301.1	Schizoid personality disorder
296.1	Bipolar
301	Personality disorders
300.4	Dysthymic disorder
300.13	Phobia
300.12	Agoraphobia, social phobia, and panic disorder
300.11	Generalized anxiety disorder
300.3	Obsessive-compulsive disorders
300.8	Acute reaction to stress
300.9	Posttraumatic stress disorder
304	Adjustment reaction
303.1	Dissociative disorder
303	Psychogenic and somatoform disorders
295	Schizophrenia and other psychotic disorders
303.4	Somatoform disorder
303.3	Psychogenic disorder
300	Anxiety, phobic and dissociative disorders
306	Other mental disorder
305.2	Eating disorder
305.21	Anorexia nervosa
302	Sexual and gender identity disorders
301.2	Antisocial/borderline personality disorder
306.9	Tension headache
297.2	Suicide or self-inflicted injury
297	Suicidal ideation or attempt

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